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标题	One-Step Catalyst-Transfer Macrocyclization: Expanding the Chemical Space of Azaparacyclophanes
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摘要（原文节选）

In this paper, we report on a one-step catalyst-transfer macrocyclization (CTM) reaction, based on the Pd-catalyzed Buchwald – Hartwig cross-coupling reaction, selectively affording only cyclic structures. This route offers a versatile and efficient approach to synthesize aza[1n]paracyclophanes (APCs) featuring diverse functionalities and lumens. The method operates at mild reaction temperatures (40 ° C) and short reaction times (2 h), delivering excellent isolated yields (>75% macrocycles) and up to 30% of a 6-me ⋯

数据表 · Table 1（24 行，含数值行 24 行）

Optimization of the Reaction Conditions for Monomer M1 to Obtain 1a

entry	precatalystb(L-Pd-G4)	ligand (L)	solvent	[M1] (M)	yield (%)c	16Nd
1	XPhos-Pd-G4	XPhos	THF	0.035	99	Y
2	RuPhos-Pd-G4	RuPhos	THF	0.035	n.d	N
3	tBu3P – Pd-G4	tBu3P	THF	0.035	77	Y
4	AmPhos-Pd-G4	AmPhos	THF	0.035	38	Y
5	tBuXPhos-Pd-G4	tBuXPhos	THF	0.035	73	Y
6	BrettPhos-Pd-G4	BrettPhos	THF	0.035	9	Ye
7	MorDalPhos-Pd-G4	MorDalPhos	THF	0.035	6	Y
8	XantPhos-Pd-G4	XantPhos	THF	0.035	n.d	N
9	PEPPSI-IPr		THF	0.035	n.d	N
10	XPhos-Pd-G4	XPhos	THF	0.070	99	Y
11	XPhos-Pd-G4	XPhos	THF	0.017	99	Y
12f	XPhos-Pd-G4	XPhos	THF	0.035	99	Y
13g	XPhos-Pd-G4	XPhos	THF	0.035	99	Y
14h	XPhos-Pd-G4	XPhos	THF	0.035	99	Y
15i	XPhos-Pd-G4	XPhos	THF	0.035	54	Y
16	XPhos-Pd-G4	XPhos	nitrobenzene	0.035	10	Y
17	XPhos-Pd-G4	XPhos	chlorobenzene	0.035	16	N
18	XPhos-Pd-G4	XPhos	1,4-dioxane	0.035	99	Y
19	XPhos-Pd-G4	XPhos	toluene	0.035	97	Y
20	XPhos-Pd-G4	XPhos	cyclohexane	0.035	81	Y
21	XPhos-Pd-G4	XPhos	THF	0.350	99	Y
22j	XPhos-Pd-G4	XPhos	THF	0.0035	20	N
23k	XPhos-Pd-G4	XPhos	THF	0.0035	60	Y
24l	XPhos-Pd-G4	XPhos	THF	0.035	81	Y

脚注（逐字）：aOptimized reaction conditions, exemplified for entry 1 and Figure 1b: (i) precatalyst: 0.04 equiv (based on M1), additional ligand: 0.04 equiv (1:1 relative to Pd), tBuONa: 2 equiv, THF, T: 40 ° C, t: 1 h (precatalyst activation), (ii) M1: 1 equiv (M/I = 25), [M1] = 0.035 M, T: 40 ° C, t: 2 h, (iii)

正文对应句（逐字）：Following reports described the synthesis of the meta congeners of aza[1n]metacyclophanes using Pd-mediated cross-coupling reaction.8,28 It was only in 2010 that the first example of an isolated hexameric APC was prepared, capitalizing on a sequential, convergent multistep synthetic approach exploit

数据表 · Table 4（14 行，含数值行 14 行）

Photophysical Properties of Selected APCs, i.e., 1, 3, 10, 11, 15, and 16, Absorption and Emission Maximum Wavelengths, Lifetime (τ), Fluorescence Quantum Yield (Φ), and Average Molar Absorption Coefficient (ε) Reported at λ absmax

entry	compound	λ absmax (nm)a	λ emmax (nm)	τ (ns)	Φ (%)	kf (ns – 1)b	knr (ns – 1)c	ε (M – 1 cm – 1)
1	16N	350	426	1.3	5	0.04	0.73	36,366
2	36N	351	423	1.3	4	0.03	0.74	56,516
3	37N	344	419	1.6	5	0.03	0.59	72,328
4	106N	389	467	3.4	39	0.11	0.18	17,844

entry	compound	λ absmax (nm)a	λ emmax (nm)	τ (ns)	Φ (%)	kf (ns − 1)b	knr (ns − 1)c	ε (M − 1 cm − 1)
5	107N	389	464	3.2	41	0.13	0.18	22,868
6	115N	363	420	1.5	65	0.43	0.23	38,440
7	116N	367	419	1.2	63	0.53	0.31	68,956
8	117N	369	418	1.1	69	0.63	0.28	108,757
9	154N	386	445	4.8	36	0.08	0.13	178,274
10	155N	389	463	2.3	49	0.21	0.22	168,859
11	156N	392	433	2.0	52	0.26	0.24	181,874
12	164N	385	446	4.5	36	0.08	0.14	131,784
13	165N	385	434	2.1	50	0.24	0.24	115,987
14	166N	393	434	1.9	49	0.26	0.27	146,294

脚注（逐字）：aIn the 15 and 16 macrocycle series, λ absmax does not correspond to the lowest electronic transitions (shoulder peaks are present).bRadiative rate constant, given by kf = Φem/ τ f.cTotal nonradiative rate constant, given by (1/ τ f) − kf.

正文对应句（逐字）：Following reports described the synthesis of the meta congeners of aza[1n]metacyclophanes using Pd-mediated cross-coupling reaction.8,28 It was only in 2010 that the first example of an isolated hexameric APC was prepared, capitalizing on a sequential, convergent multistep synthetic approach exploit

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